

Total No. of Questions : 4]

SEAT No. :

**PA-10263**

[Total No. of Pages : 1

**[6010] 148**

**B.E. (Mechanical/Automobile Engineering) (In Sem)**

**Honors in “Electric Vehicle”**

**E-VEHICLE STANDARDS, CHARGING & SAFETY**

**(2019 Pattern) (Semester - VIII) (302036 MJ)**

*Time : 1 Hour]*

*[Max. Marks : 30*

**Instructions to the candidates:**

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1) a)** What are the Electrical safety Requirements of a vehicle. [5]  
b) Explain the electric power train. [5]  
c) Write a short notes on Protection against electric shock. [5]

OR

- Q2) a)** Explain the Measurement of Electrical Energy Consumption. [5]  
b) Explain A Rechargeable Electrical Energy Storage System. [5]  
c) Explain the Electric Power Train Vehicles-Construction and Functional Safety Requirements. [5]

- Q3) a)** What are the Requirements for Regenerative Braking System. [5]  
b) Explain EV integration in the power grid. [5]  
c) Write a short notes on charging points of electric vehicle. [5]

OR

- Q4) a)** Explain the Electric Vehicle Battery Fast Charging and discharging. [5]  
b) Write a short notes on Battery Performance for electric vehicle. [5]  
c) Explain role of EV aggregators in EVGI (Electric Vehicle Grid Integration) framework. [5]

